Chapter 6
Evaluation of the Uncertainty in Indirect Measurements



6.1 Uncertainty Propagation Law

Consider a greatness W described by the function $W = f(a, b, c \ldots)$, where $a, b, c, \ldots$ are statistically independent[1] variables. If the most likely values for these quantities are $\overline{a}, \overline{b}, \overline{c}, \ldots,$ the most likely value for W will be $W = f\left(\overline{a}, \overline{b}, \overline{c}, \ldots\right)$. Expanding the W function in the Taylor series, we will have:

$$\begin{aligned} W_i \approx W\left(\overline{a}, \overline{b}, \overline{c} \ldots\right) + \frac{\partial W}{\partial a}\left(a_i - \overline{a}\right) + \frac{\partial W}{\partial b}\left(b_i - \overline{b}\right) + \frac{\partial W}{\partial c}\left(c_i - \overline{c}\right) + \ldots \\ + \frac{1}{2}\frac{\partial^2 W}{\partial a^2}\left(a_i - \overline{a}\right)^2 + + \frac{1}{2}\frac{\partial^2 W}{\partial b^2}\left(b_i - \overline{b}\right)^2 + \frac{1}{2}\frac{\partial^2 W}{\partial c^2}\left(c_i - \overline{c}\right)^2 + \ldots \end{aligned} \tag{6.1}$$

We will make an approximation disregarding the quadratic terms when $\left(a_i - \overline{a}\right)$, $\left(b_i - \overline{b}\right)$, $\left(c_i - \overline{c}\right)\ldots$ are of the order of greatness of the standard deviation σ_a, σ_b, σ_c, … Then, $\sigma_a = \left(a_i - \overline{a}\right)$ and $\frac{1}{2}\frac{\partial^2 W}{\partial a^2}\left(a_i - \overline{a}\right)^2 \approx 0$.

This condition applies to all other variables.

Therefore, we have:

$$W_i - \overline{W\left(\overline{a}, \overline{b}, \overline{c} \ldots\right)} = \Delta W_i = \left[\frac{\partial W}{\partial a} \cdot \sigma_a + \frac{\partial W}{\partial b} \cdot \sigma_b + \frac{\partial W}{\partial c} \cdot \sigma_c + \ldots\right] \tag{6.2}$$

The term $\frac{\partial W}{\partial a}$ represents the partial derivative of W concerning the variable a, $a = \overline{a}$, calculated in which all other variables were kept constant.

[1] The variables are considered statistically independent when the variation of one does not influence the variation of the other, that is, all behave in a detached manner. Statistically, these variables have a correlation coefficient equal to zero.

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Knowing a Little More ... (Fig. 6.1) (continued)
son. After this loss, Brook returned to live with his father. In 1725, he married again, with his father's approval. The chosen one was Sabetta Sawbridge of Olantigh. In 1729, after his father's death, Brook inherited the property. However, personal tragedies continued to torment him when his second wife died in childbirth. The child, named Elizabeth, managed to survive. Taylor lived a few more years (died at age 46), but his mathematical deeds are surprising and probably not deepened due to personal factors (disappointments, fragile health). It gave rise to a new branch in mathematics called "Calculus of finite differences," parts integration, and the series known as Taylor's expansion.

6.2 When Variables Are Statistically Dependent

Statistically, dependent variables behave in a linked manner; that is, the variation of one influences the variation of another. These variables have a different correlation coefficient from zero.

In the presence of statistically dependent variables, measurement uncertainty should consider the correlation coefficient (r) between variables. The correlation coefficient may vary between [−1, +1], being zero when the variables are independent, and their expression is given by:

$$r = \frac{s_{ab}}{s_a s_b} \tag{6.4}$$

$$r(a,b) = \frac{\sum_{i=1}^{n}(a_i - \overline{a})(b_i - \overline{b})}{\sqrt{\sum (a_i - \overline{a})^2}.\sqrt{\sum (b_i - \overline{b})^2}} \tag{6.5}$$

Where s_a and s_b are the standard deviation of a and b variables, and s_{ab} is the standard deviation of the correlation between the variables given by Eq. (6.5).

If the variables are dependent on each other, the uncertainty propagation equation will be:

$$u_W^2 = \left(\frac{\partial W}{\partial a}\right)^2 u_a^2 + \left(\frac{\partial W}{\partial b}\right)^2 u_b^2 + \left(\frac{\partial W}{\partial c}\right)^2 u_c^2 + \ldots + 2\left(\frac{\partial W}{\partial a}\right) \times \left(\frac{\partial W}{\partial b}\right) r(a,b) u_a u_b + 2\left(\frac{\partial W}{\partial a}\right)\left(\frac{\partial W}{\partial c}\right) r(a,c) u_a u_c + \ldots \tag{6.6}$$

where $r(a, b)$, $r(a, c)$, $r(b, c)$, ... are the correlation coefficients between the variables (a, b, c...).

Table 6.1 Relative uncertainties of some functions

Case 1	$F = x \cdot y$	$u_F = F \cdot \sqrt{\left(\frac{u_x}{x}\right)^2 + \left(\frac{u_y}{y}\right)^2}$
Case 2	$F = \frac{x}{y}$	$u_F = F \cdot \sqrt{\left(\frac{u_x}{x}\right)^2 + \left(\frac{u_y}{y}\right)^2}$
Case 3	$F = x^m$	$u_F = F\left(m\frac{u_x}{x}\right)$
Case 4	$F = x^P + y^q$	$u_F = \quad F\sqrt{\left(p\frac{u_x}{x}\right)^2 + \left(q\frac{u_y}{y}\right)^2}$
Case 5	$F = \log x$	$u_F = 0,43429 \cdot \left(\frac{u_x}{x}\right)$
Case 6	$F = \ln x$	$u_F = \left(\frac{u_x}{x}\right)$
Case 7	$F = e^x$	$u_F = F \cdot u_x$
Case 8	$F = 10^x$	$u_F = F \cdot (2.3026 \cdot u_x)$

Note: u_F is the combined uncertainty of the function F

6.3 Method of Relative Uncertainties[2]

If we do not want to calculate the partial derivative of a function or this knowledge has not yet been addressed by the reader, follow, in Table 6.1, a propagation relationship of uncertainty in which the relative uncertainties are used in some mathematical functions.

The relationship between the measurement uncertainty of a variable x and the value of this variable determines relative measurement uncertainty.

The variables must be statistically independent.

$$u_{\text{relative}} = \frac{u_x}{x} \tag{6.7}$$

Solved Exercise 6.1

Deduces the formula of relative uncertainty to case 1: $F = x \cdot y$.

Solution

- The partial derivative of function F concerning x: $\frac{\partial F}{\partial x} = y$.
- The partial derivative of function F concerning y: $\frac{\partial F}{\partial y} = x$.

$$u_F^2 = \left(\frac{\partial F}{\partial x}\right)^2 u^2{}_x + \left(\frac{\partial F}{\partial y}\right)^2 u^2{}_y$$

$$u_F^2 = (y)^2 u^2{}_x + (x)^2 u^2{}_y$$

[2]This method can be employed when derivation techniques are not known. It can be applied to studies on the calculation of measurement uncertainties at the high school/technical level.

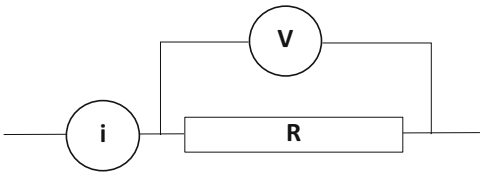


Fig. 6.2 Measurement circuit

$$U_R = k.u_R = 2.00 \times 0.3256 = 0.6513 \ \Omega$$
$$U_R = 0.7 \ \Omega$$

The result will be: $R = (52.4 \pm 0.7)\ \Omega$

6.4 Evaluation of the Effective Degree of Freedom for Relative Uncertainties

In Chap. 5, we saw that we must use the Welch-Satterthwaite Eq. (5.6) or Eq. (5.7) to determine the effective degree of freedom.

In the case of indirect measurements, we must apply the uncertainties relative to Eq. (5.6), since the quantities involved have different units. In this way, the Welch-Satterthwaite equation is like this:

$$\frac{u_{Rc}^4}{\upsilon_{\text{eff}}} = \frac{u_{R1}^4}{\upsilon_1} + \frac{u_{R2}^4}{\upsilon_2} + \ldots + \frac{u_{Ri}^4}{\upsilon_i} \tag{6.8}$$

- u_{Rc} is the combined relative uncertainty of the quantity that we need to determine the degree of freedom.
- u_{R1}, u_{R2}, u_{R3}, ..., u_{Ri} are the standard relative uncertainties of each i source (Type A and B uncertainties).
- υ_1, υ_2, υ_3, υ_i … are the degrees of freedom of each i source of uncertainties;
- υ_{eff} is the effective degree of freedom associated with standard combined relative uncertainty.

Solved Exercise 6.4

A cylindrical metal bar (Fig. 6.3) has a diameter $d = (2.50 \pm 0.01)$ cm, $k = 2.37$, and 95.45%; length $L = (30.48 \pm 0.01)$ cm, $k = 2.28$, and 95.45%; mass $M = (1158.0 \pm 0.1)$ g, $k = 2.23$, and 95.45%. Whereas the equation calculates the volume of a cylinder:



Fig. 6.3 Cylindrical bar

6.5 Sensitivity Coefficient

Considering a function W (a, b, c, …), in which a, b, c, … are its variables, as described in Sect. 6.1, we have Eq. (6.3) below.

$$u_W^2 = \left(\frac{\partial W}{\partial a}\right)^2 \cdot u_a^2 + \left(\frac{\partial W}{\partial b}\right)^2 \cdot u_b^2 + \left(\frac{\partial W}{\partial c}\right)^2 \cdot u_c^2 + \ldots$$

This equation describes the propagation of uncertainties of a function W.

Partial derivatives $\left(\frac{\partial W}{\partial a}\right)$ $\left(\frac{\partial W}{\partial b}\right)$ $\left(\frac{\partial W}{\partial c}\right)$ … describe the variation of function W concerning each variable a, b, c, …

Important
In metrology, these partial derivatives are called sensitivity coefficient (SC_i) and describe how each input a, b, c,... influences the output value W.

Experimentally, if all variables remain constant and only one, for example, the variable a, changes, we can verify the variation of W. In Solved Exercise 6.4, we saw that the partial derivative method allows us to determine the function's sensitivity coefficient concerning each variable.

The sensitivity coefficient

$$\frac{\partial \rho}{\partial M} = \frac{4}{\pi d^2 L}$$

indicates the variation of cylinder density as a function of the variation of its mass (M).

The same happens with the expression

$$\frac{\partial \rho}{\partial d} = \frac{-8M}{\pi d^3 L}$$

which indicates the change of cylinder density by variation of its diameter (d), and the expression

$$\frac{\partial \rho}{\partial L} = \frac{-4M}{\pi d^2 L^2}$$

indicates the variation of cylinder density as a function of the variation of its length (L).

Knowledge of the sensitivity coefficient is essential to know how much a variable influences the result of an indirect measurement, minimizing its influence and, thus, its measurement uncertainty.

In Solved Exercise 6.4, we saw that the largest sensitivity coefficient (in absolute value) of the three-cylinder variables in question—diameter (d), mass (M), and length (L)—is the sensitivity coefficient of cylinder density concerning its diameter:

$$\frac{\partial \rho}{\partial M} = \frac{4}{\pi d^2 L} = 0.006684/\text{cm}^3$$

$$\frac{\partial \rho}{\partial d} = \frac{-8M}{\pi d^3 L} = -6.19174\ \text{g/cm}^4$$

$$\frac{\partial \rho}{\partial L} = \frac{-4M}{\pi d^2 L^2} = -0.25393\ \text{g/cm}^4$$

This fact indicates that we should be more concerned with diameter measurement uncertainty (higher absolute value of SC_i), if we want to minimize its influence on the result of cylinder density.

6.5.1 Sensitivity Coefficient Transforming Uncertainties

The sensitivity coefficient is also applicable, when we want to transform a measurement uncertainty that presents itself into a quantity to another quantity. This case is widespread when we want to measure a quantity, and the instrument used provides a sign in another quantity. We can cite sensors (thermoresistors and thermocouples) and pressure and temperature transducers.

Take as an example the PT-100 platinum resistance thermometer (RTD), which is widely used in industry and thermometry laboratories, because it has low uncertainty and good accuracy. It measures temperature through the variation of its platinum resistance, which, at zero °C, has a value close to 100 ohms.

The Callendar–Van Dusen equation describes the relationship between resistance (R) and temperature (T) of platinum resistance thermometers.

$$R(T) = R_0 \left(1 + AT + BT^2 - 100CT^3 + CT^4\right) \tag{6.9}$$

where $R(T)$ is the resistance value at the desired temperature T, and R_0 is the value of platinum resistance at 0 °C. A, B, and C are the RTD coefficients, with typical values for an industrial platinum resistance thermometer such as:

$$A = 3.9083 \times 10^{-3}/^\circ\text{C}; \quad B = -5.775 \times 10^{-7}/^\circ\text{C}^2;$$

$$C = 0 \ \ (T \leq 650^\circ C); \quad C = -4.183 \times 10^{-12}/^\circ C^4 \ \ (T > 650^\circ C)$$

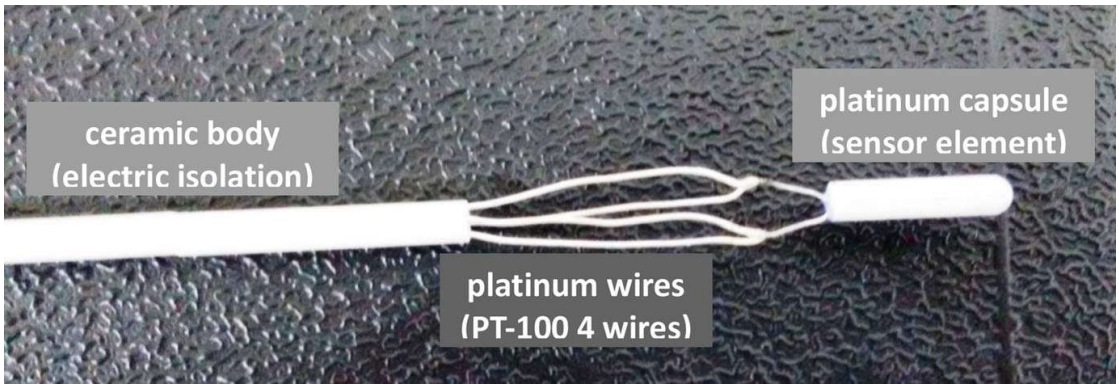


Fig. 6.4 Four-wire platinum resistance thermometer. (Photo by the authors)

degree of purity and encapsulated in ceramic or glass bulbs (Fig. 6.4). Several types of resistance thermometers, from the standard thermometer to the more robust industrial thermometer, may have uncertainties in the tenth of the degree. The most common platinum types are those with a resistance of 25 ohms, 100 ohms, 500 ohms, or 1000 ohms at the ice point (0 °C) (Fig. 6.4).

A two-, three-, or four-wire resistance bulb connection is used according to the type of instrument and the desired accuracy in the measurement. The four-wire thermometers are the most exact and are called half standard. The PT-25 is considered standard, being the most accurate and uncertain of 0.001 Ω. Its main constructive characteristics are:

(i) The sensor element is made of platinum with a purity better than 99.999 %;
(ii) Great thermometer stability and measurement accuracy, with uncertainty values of (0.0006 to 0.01) °C.

6.6 Proposed Exercises

6.6.1 The expression gives the density of a sphere:

$$\rho = \frac{M}{V} = \frac{6M}{\pi D^3}$$

Consider $M = (1000 \pm 1)$ g, $D = (8.000 \pm 0.002)$ cm, and $k = 2.00$ and 95.45 %. Determine the density and the measure uncertainty using: